COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 6 — Prototype Model

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Section 11: Classroom Activity — Prototype a Library Search

Rapid low-fidelity prototyping and usability testing for a campus library mobile search and reservation workflow.

1. Prototype Screen Specifications (Wireframe Architecture)

- **Screen 1 (Search & Results View):** Header: "Campus Library Catalog". Centered input search bar with live auto-complete ("Title, Author, Subject"). Results list displaying book cover thumbnail, Title, Author, Edition year, Call number, and dynamic badge: [Available: 3 Copies] in green or [Reserved / Borrowed] in red.
- **Screen 2 (Book Details View):** Expanded title metadata: Abstract, Publisher, ISBN-13, physical shelf location ("Floor 2, Shelf B-14"). Availability status breakdown: 2 available in Central Library, 1 in Departmental Annex. Prominent primary CTA button: [Reserve This Copy].
- **Screen 3 (Reservation Confirmation):** Modal summary: "Reservation Successful!". Displays unique 6-digit Reservation Code, Pickup Location Desk, and countdown notice: "Please collect physical copy within 48 hours before reservation lapses." Buttons: [Add to Calendar] and [Back to Search].

2. Usability Walkthrough & Partner Observations

- **Test Task:** "Find an available copy of Software Engineering by Pressman and reserve it."
- **User Hesitation 1:** Partner hesitated on Screen 1 because the search filter dropdown default was set to "ISBN" rather than "All Fields", causing initial search by title to fail.
- **User Hesitation 2:** On Screen 2, the user was confused whether "Reserve" meant checking out a digital PDF or holding a physical book at the physical circulation desk.
- **Required Prototype Revisions:** (a) Set default search scope to global text match; (b) Add clear copy label: "Physical Copy Hold at Circulation Counter"; (c) Display pickup expiry date prominently on confirmation screen.

Section 14: Homework — Prototype Plan for Online Food Ordering App

Structured test plan to evaluate user experience during the restaurant cart checkout and tip allocation flow.

- **1. Core Uncertainty Investigated:** Will users understand dynamic surge pricing delivery fees and customizable driver tip toggles during checkout, or will fee ambiguity cause cart abandonment?
- **2. Target Review Users:** Sample cohort of 15 university students and 10 campus faculty members who regularly order lunch online.

- **3. Represented Behaviors & Screens:** Interactive clickable wireframes representing: (a) Restaurant menu dish modifier modal; (b) Itemized cart review with breakdown of item totals, platform fee, taxes, and surge pricing; (c) Driver tip selector (Rs. 20, Rs. 30, Custom).
- **4. Three Specific Evaluation Tasks:** (a) Customize a burger with extra cheese and add to cart; (b) Identify the surge delivery charge on the checkout bill; (c) Remove an item from cart and complete simulated checkout.
- **5. Feedback Capture & Decision Framework:** Record screen interactions, task completion time, and hesitation points. If more than 20% of users fail to locate the itemized tax/surge tooltip within 5 seconds, redesign the billing summary into an expandable drawer before writing production frontend code.

DAKSH